

Reasons for Enucleation after Plaque Radiotherapy for Posterior Uveal Melanoma

Clinical Findings

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Abstract: A review of 1019 patients with posterior uveal melanomas who were treated with episcleral plaque radiotherapy showed that 59 (6%) required enucleation of the affected eye. Seventeen of the enucleations were done within 1 year, 30 within 2 years, and 55 within 5 years. The enucleated eyes had been treated with a cobalt-60 plaque in 39 cases, a ruthenium-106 plaque in 13 cases, an iodine-125 plaque in four cases, and an iridium-192 plaque in three cases. The primary reasons for enucleation included tumor regrowth in 30 cases (51%), neovascular glaucoma in 18 (31%), patient request in five (8%), scleral melting in four (7%), painful bullous keratopathy in one (2%), and hemolytic glaucoma in one (2%). The time interval from plaque treatment to eventual enucleation averaged 29 months. Tumor regrowth requiring enucleation was detected clinically an average of 28 months after treatment. In these cases the average radiation dose to the tumor apex was 7700 cGy and to the tumor base 36,000 cGy. Uncontrollable neovascular glaucoma occurred an average of 38 months after plaque radiotherapy and, most commonly, after cobalt-60 plaques were used. Eyes with plaque-induced scleral melting eventually required enucleation after an average of 10 months. In all cases of scleral melting, the tumor was ciliochoroidal. *Ophthalmology* 96:919-924, 1989

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Episcleral plaque radiotherapy has been used in the treatment of posterior uveal melanoma for over 20 years and is an effective method of controlling this tumor locally.¹⁻¹⁰ Successful clinical regression of the tumor is observed in most cases but, unfortunately, treatment complications do occur.¹¹⁻¹⁶ In most cases that develop complications of radiotherapy, the eye is retained with poor vision.¹⁶ However, in some cases, the complications are severe and require enucleation of the irradiated eye.^{4,5,9,13,17} In 1977 MacFaul and Morgan¹⁷ evaluated a series of 100 eyes with choroidal melanoma treated with cobalt plaque applicators at Moorfields Eye Hospital between 1960 and 1975. They found that 23 eyes eventually required enucleation. Since their study, there have been no reports of large series evaluating the clinicopathologic